Gun Game — Unreal Hitscan Prototype

Source text companion

Text export from the original coursework submission. Original wording and citations are retained; layout, figures and formatting are not reproduced. This is not a newly reviewed or replicated study.

Gun Game Team members: Airron Kennedy , TABLE OF CONTENTS 01 Player Design 02 Gun Design 03 Timer UI Design 04 Target Design The Default Basic unreal blueprints for player character were used to move and fire a gun that can be picked up. Only think that was changed was that a UI Widget will appear when you start the game! 01 Player Design The gun is a modified version of the default unreal gun blueprint. Removing the projectile and changing to a line trace allows me to create a “Hitscan” effect where targets are destroyed when the player fires at them 02 Gun Design https://www.youtube.com/watch?v=IT6pYhqhNyM (How To Create Automatic Guns in Unreal Engine 5 | Line Trace Tutorial) https://www.youtube.com/watch?v=5hTpbumwLE0&t=61s (Unreal Engine 4 - Line Trace Full Auto Weapon) 02 Gun Design Research methods The timer UI was set up to allow the player 20 seconds to destroy all the targets on the screen, upon hitting the last target, the game is paused where you can see how much time is remaining for personal record. This design counts down milliseconds then after it reaches 0, it takes a second off of the seconds counter. 03 Timer UI Design 03 Timer UI Design Cont https://www.youtube.com/watch?v=nHK97x6ILYo (Countdown Timer | End Game Screen - Unreal Engine 4 Tutorial) 03 Timer UI Design Research methods The targets were designed in a way to be able to receive damage from the line trace gun. Simultaneously , they are tracking the player position so when the player is looking directly at them, they are lit up a different color so the player is indicated that they are aiming at them 04 Target Design 04 Target Design Cont https://www.youtube.com/watch?v=GXcGAvik2cc&t=750s (Display Name On Screen Of Object You Are Looking At - Unreal Engine Tutorial) https://www.youtube.com/watch?v=mqhPIlx4jdc&t=371s (UE4 - Player Tracking, Find Look Rotation Node in Blueprints) https://www.youtube.com/watch?v=3OCisqQMaVc&t=856s (Practice Shooting Range | Moving Targets | Score In Widget - Unreal Engine 4 Tutorial) 04 Target Design Research methods